ML Assignment 5

UNIVERSITY OF HARTFORD

Assignment 5

K-means Clustering

Student: John Bennett

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1 Objective

The goal of this assignment is to implement k-means clustering algorithm and compare its efficacy through the following tasks:

- Implement k-means clustering algorithm
- Validate clustering results using sum of square errors distortion measure
- Use elbow method to determine an optimal K
- Compare implementation to Python's Sci-kit Learn's implementation

2 Data Set Overview and Cleaning

The data set explored in this assignment was a synthetic dataset with characteristics to support the development of the K-means clustering implementation. It consists of two features that define its 5000 data objects within a feature space. The data objects have relative relationships (similar positions within the feature space) that lend itself to clustering algorithms. Since the data was synthetic, no cleaning was performed.

3 Approach

K-means is one of several machine learning clustering algorithms that use an unsupervised approaches to group data together without explicit classification models. The goal of these algorithms are to find associations within data rather than generate a model to predict future data. Where data objects and their feature values are proxies for natural entities (like, genes, customers, or celestial bodies), clustering algorithms can serve to demonstrate relationships between them. Clustering also supports identifying what features can be used to effectively group these entities together in meaningful ways. Each of these algorithms have both advantages and restrictions that effect their individual ability to meaningfully group entities together. Ideally, the data found within a cluster is very similar to other members of that cluster while also being different from the members of other clusters.

K-means clustering has the characteristic of being centroid-based, with implications for the topology that is possible for its clusters to have. In this approach, a hyper-parameter of k, indicating the number centroids, is provided to the algorithm and each object in the dataset upon which clustering analysis is performed is ultimately assigned to a cluster based on its relative proximity to one of the centroids produced by the algorithm. The centroid itself consists of a set feature values that define a position within the feature space defined by the dataset being analyzed. It can be thought of as a positional mean that places it at the center of its cluster. The topological implication of the centroid approach to defining data clusters is that each cluster is a hyper spherical shape only capable of convexity with a constant boundary defined by the farthest proximity of the object assigned to it.

The K-means algorithm works by randomly assigning "k" data points as initial centroids. This is likely the only time in the algorithm that a centroid will have the same

feature value set as an object in the dataset. The algorithm proceeds by iteratively assigning each of the data objects defined by the dataset to one of the centroids to establish k clusters and then updates the centroid for that cluster as the set of mean feature values with respect to all of the cluster's data objects. Then the algorithm re-assigns each object based on this updated set of centroids. It proceeds in this fashion until the change in distance between the centroids across sequential iterations is less than a predetermined minimum threshold (or until a maximum number of iterations is reached).

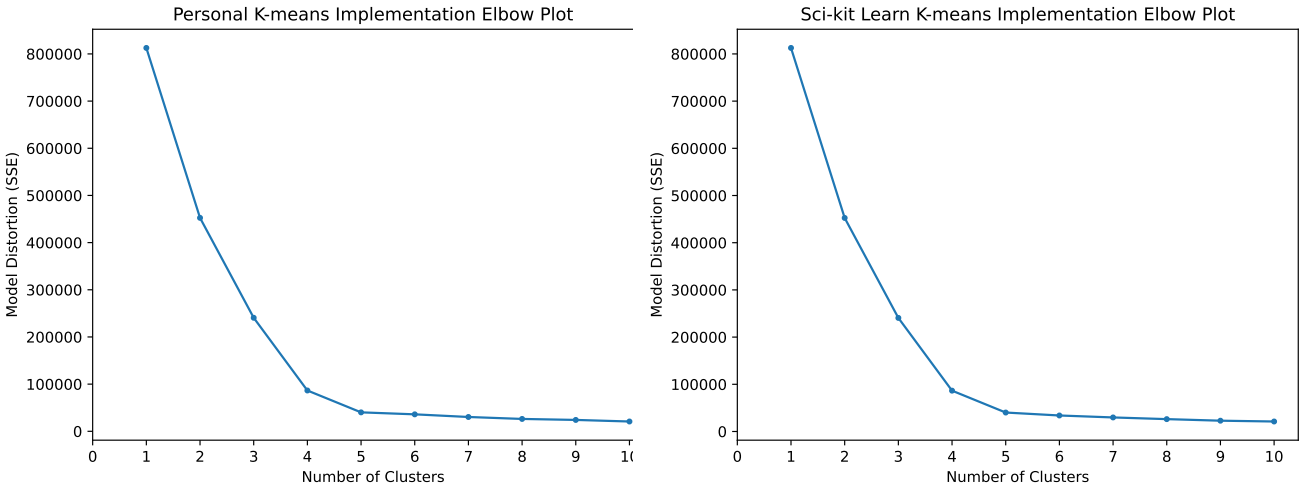
The fact that the number of clusters is predetermined is addressed by deploying the algorithm over a set of potential k values. The reason that this is necessary is that the natural number of clusters in a dataset is rarely identifiable prior to clustering. To determine what is the most natural number of clusters for a dataset, a total distortion statistic is collected for the clustering results collected for each candidate k-value. In an approach referred to as the "elbow method," a bi-variate line plot is (typically) constructed between the k-values and their respective distortion statistic value. The elbow of this plot determines the best k value and is identifiable as the k-value where the change in distortion is most significantly diminished in comparison the next largest k-value. This can also be achieved using a discrete implementation of the second derivative of the distortion plot so that the location where the elbow occurs can be established computationally. In this implementation of k-means, sum of squared errors (SSE) between the individual clusters is totaled to define the total distortion for a given k-value, this approach accounts for the similarity between objects within the same cluster; there are other approaches to validating k-values that also account for distances between members of different clusters (referred to as separation).

To achieve a smooth distortion plot the implementation of the k-means algorithm will generate multiple candidate models for each k-values, and choose the one with the best distortion score. This adjusts for the randomness in the clustering algorithm in identifying centroids. By creating multiple models with random initial centroids, the implementation accommodates bad clustering results due to unfortunate initial cluster values.

4 Results and Discussion

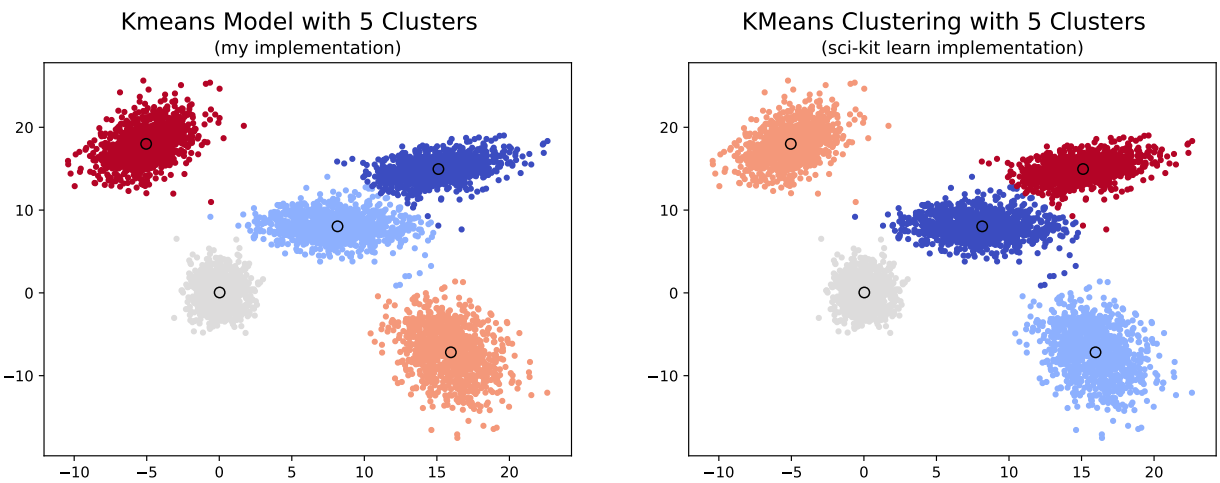
The implementation of k-means was successful generating nearly identical results with python's Sci-kit Learn implementation of k-means. Though time complexity was not tracked between the two implementations, the only exception to equal performance for this assignment's dataset is the relative time complexity due to the fact that the custom implementation is built using python's Numpy library.

Figure 1: Comparative Elbow Plots between K-means Implementations



The elbow plots in Figure 1 show a significant amount of similarity between results for the range of k-values tested for optimization. In both plots, it is clear that the best k-value for this dataset is $k = 5$. There is only a slight difference in the subsequent k-values most clearly seen at $k = 6$.

Figure 2: Comparative Scatter Plots with Clusters between K-means Implementations



The scatter plots in Figure 2 show a the same results between the two implementations where they only differ in the coloration of each cluster label, which is arbitrary. The identical results seen in the scatter plots are verified by the adjusted rand index (ARI) and the total SSE. The ARI establishes the level of agreement between the cluster labels of each model. Its value is between 0 and 1 where one indicates that all pairs of data points are labeled the same or differently between the label sets of the two implementations. In this case, the ARI value was 1.0 indicating perfect alignment. Additionally, the total SSE was 40158.49 for both models.

The comparison between implementations indicates that the personal implementation of the k-means clustering algorithm was successful. beside the implementation,

this assignment provided the opportunity to practice techniques for comparison between clustering results. These techniques are a valuable part of validating clustering results, both between different sets of hyperparameters for the same algorithm and, in other contexts, differences between clustering results using diferent algorithms.